

Supplemental information: Nurture over nature in fate decisions of antigen-specific CD8+ T cell clones from an endogenous repertoire

STAR METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER RRID
Antibodies		
murine		
aCD4 PE-Dazzle 594	BioLegend	AB 2563684
aCD19 PE-Dazzle 594	BioLegend	AB 2564000
aCD8a BV785	BioLegend	AB 2562610
aCD44 FITC	BioLegend	AB 493685
Murine DNA barcodes		
TotalSeq™-C0301 anti-mouse Hashtag 1	BioLegend	AB 2800693
TotalSeq™-C0302 anti-mouse Hashtag 2	BioLegend	AB 2800694
TotalSeq™-C0303 anti-mouse Hashtag 3	BioLegend	AB 2800695
TotalSeq™-C0304 anti-mouse Hashtag 4	BioLegend	AB 2800696
TotalSeq™-C0305 anti-mouse Hashtag 5	BioLegend	AB 2800697
TotalSeq™-C0306 anti-mouse Hashtag 6	BioLegend	AB 2819910
TotalSeq™-C0307 anti-mouse Hashtag 7	BioLegend	AB 2819911
TotalSeq™-C0308 anti-mouse Hashtag 8	BioLegend	AB 2819912
TotalSeq™-C0309 anti-mouse Hashtag 9	BioLegend	AB 2819913
TotalSeq™-C0310 anti-mouse Hashtag 10	BioLegend	AB 2819914
TotalSeq™-C0075 anti-mouse CD90.2	BioLegend	AB 2832346
TotalSeq™-C0182 anti-mouse CD3	BioLegend	AB 2832261
Bacterial Strains		
Listeria monocytogenes-SIINFEKL	Zehn et al. ¹	
Chemicals, Peptides, Enzymes		
Propidium Iodide (PI)	Life Technologies	Cat#P1304MP
Experimental Models: Organisms/Strains		
C57Bl/6J0laHsd	Envigo	Stock number: 057
MHC multimer		
H-2K ^b /mβ2m/Ova 257-264 Streptavidin-APC/BV421	In house	
Software and Algorithms		
FloJo V10	FlowJo LLC	
Prism 9	Graphpad	
Microsoft Excel	Microsoft	
Python 3.11.11	Python Software Foundation	
Scanpy 1.11.0	https://scanpy.readthedocs.io/en/stable/	
Scirpy 0.12.2	https://icbi-lab.github.io/scirpy/	
Matplotlib 3.10.1	https://matplotlib.org/	
Seaborn 0.13.2	https://seaborn.pydata.org/	
Anndata 0.11.3	https://anndata.readthedocs.io/en/latest/	
R 4.4.3	https://www.r-project.org/	
Brms 2.22.0	https://cran.r-project.org/web/packages/brms/index.html	

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be managed by the lead contact Dirk Busch (dirk.busch@tum.de).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All data and supplementary information generated or analyzed during this study are included in this article. This paper does not report original code. Any additional information and notebooks containing all steps of data processing and analysis of single-cell RNA sequencing reported in this paper are available on GitHub (<https://github.com/AdrianStraub/github-repo-Straub-et-al-2025>). Any additional information or raw data are available from the lead contact upon reasonable request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

C57BL/6J OlaHsd (females, 6-10 weeks) were purchased from Envigo. All animal experiments were approved by the district government of upper Bavaria (Department 5 – Environment, Health and Consumer Protection).

METHOD DETAILS

Infections

Mice were infected i.v. with 5000 colony-forming units (CFU) of recombinant *Listeria monocytogenes* expressing Ovalbumin¹.

Peptide-MHC (pMHC) multimer staining

Biotinylated pMHC molecules for the generation of pMHC multimers were refolded according to the protocol as previously described². 0,4 µg of biotinylated pMHC I molecule, 0,5 µg of Streptavidin-APC or Streptavidin-BV421 and 50 µl of FACS buffer for every 5×10^6 cells, were preincubated for at least 30 min for multimerization. Cells were then incubated with the multimer mix for 40 min. 20 min before the end of the staining period antibodies for the staining of surface antigens were added. PI for live/dead staining was added 5 min before the end of the staining period.

Cell sorting for single-cell RNA sequencing

Spleens were collected and mashed through a 70-µm cell strainer to generate a single-cell suspension. Splenocytes, following red blood cell lysis, were pre-stained for 20 min with two independent H-2K^b/SIINFEKL-multimers (Streptavidin-APC and Streptavidin-BV-421) and subsequently stained for CD4, CD19, CD8 for further 20 min at 4 °C in the dark (Multimer staining see above). After washing with FACS buffer (PBS with 0.5% BSA and 2 mM EDTA), the cells were stained with PI for 5 min and washed again. Individual donors were labelled with a unique TotalSeq™-C anti-mouse Hashtag antibody combination according to the manufacturer's instructions. 20 individually labeled donors were pooled for cell sorting and a total amount of 45.000 cells were sorted for CD4⁺, CD19⁺, CD8⁺, CD44^{hi}, H-2K^b/SIINFEKL-Streptavidin-APC⁺, H-2K^b/SIINFEKL-Streptavidin-BV421⁺ on a MoFlo Astrios cell sorter (Beckman Coulter) into a single 96well for downstream processing of the 10X genomics library preparation. For retrospective analysis of flow cytometry data, FlowJo software (FlowJo LLC) was used.

Single-cell RNA sequencing

After cells have been sorted, they were centrifuged and the supernatant was carefully removed. Cells were resuspended in the Mastermix + 37.8 µl of water before 70 µl of the cell suspension were transferred to the chip. (Step 1.1 and 1.2 of the original 10X genomics protocol). After each step, the integrity of the pellet was checked under the microscope to ensure that all cells are loaded onto the chip. From here on, 10 experiments have been performed according to the manufacturer's protocol

(Chromium next GEM Single Cell VDJ V1.1 with Feature Barcode, Rev D). QC has been performed with a High sensitivity DNA Kit on a Bioanalyzer 2100 as recommended in the protocol and libraries were quantified with the Qubit dsDNA hs assay kit. All steps have been performed using RPT filter tips and DNA LoBind tubes.

scRNA seq data analysis

References GRCh38-2020-A and vdj_GRCh38_alts_ensembl-5.0.0 were used for Transcriptome and VDJ annotation (CellRanger 6.0.2), respectively. Data preprocessing has been performed according to published current best practices in scRNA sequencing analysis³. Data analysis was performed with SCANPY V1.11.10⁴ and Scirpy 0.12.2⁵. Briefly, samples were demultiplexed using HashSolo⁶ included in SCANPY. Gene expression counts were corrected for ambient mRNA using SoupX⁷. Cells with less than 200 genes as well as genes present in less than 20 cells were excluded. Variable TCR genes ('Trav', 'Traj', 'Trbv', 'Trbj', 'Trbd') as well as ribosomal ('Rpl', 'Rps') and histone genes ('Hist') were removed prior to dimensionality reduction and clustering to avoid T cell clonotype bias (TCR genes) and cell cycle bias (ribosomal and histone genes) in clustering analysis. Counts were normalized per cell and logarithmized. Highly variable genes were identified via scanpy's inbuilt function `pp.highly_variable_genes` (flavor = 'seurat_v3') and the top 5000 most variable genes were selected for further dimensionality reduction and data analysis. Individual sequencing runs were batch corrected using scvi's⁸ graph based deep learning method as described in current single cell best practices³. Analysis notebooks containing raw data outputs are openly accessible under (<https://github.com/AdrianStraub/github-repo-Straub-et-al-2025>).

Clonotype definition from single-cell RNA sequencing data

Clonotype analysis was performed using Scirpy⁵. Cells belonging to one clonotype were defined to have identical α and β chain amino acid sequences. Only primary pairs of TRA/TRB chains were considered when multiple chains were detected.

Normalization of clone sizes across donors

To account for sample size variation across mice, we performed a size factor normalization for clone sizes. First, we calculated the size factor for each donor mouse by dividing the mean number of analyzed epitope-specific T cells across all donors with the number of epitope-specific T cells analyzed in the respective donor. To then calculate the normalized clone size, we multiplied the measured clone size with the size factor of the corresponding donor and further normalized the data so that the smallest normalized clone size yielded 1.

- 1) $SF_i = S_m / S_i$
- 2) $N_i' = C_i \times SF_i$
- 3) $N_i = N_i' / N_{\min}'$, where

SF_i = Size factor (in donor i)

S_m = mean number of analyzed epitope-specific T cells (across all donors)

S_i = number of analyzed epitope-specific T cells (in donor i)

C_i = clone size (in donor i)

N_i = normalized clone size (in donor i)

$N_{\min}$ = smallest normalized clone size

QUANTIFICATION AND STATISTICAL ANALYSES

General statistical analyses

General statistical analyses were performed using the GraphPad PRISM software and R. Statistical tests were used as indicated in the figure legends

Variance analysis and Bayesian modeling

Variance analysis was conducted on two features: the normalized clone size and the TE/MP enrichment ratio. Analogous to Plambeck et al.⁹, total variance was calculated as the variance of feature values

between all clones. TCR-extrinsic variance was calculated as the mean of the variances of the feature values in all clonotypes. TCR-intrinsic variance was calculated as the variance of the mean feature values in all clonotypes.

To investigate the relative contributions of TCR intrinsic and TCR extrinsic factors, we employed a Bayesian hierarchical modeling framework implemented using the brms package (v2.22.0)¹⁰ in R (v4.4.3). The brms package is a high-level interface to the Stan probabilistic programming language, enabling the specification of generalized multilevel models (linear and non-linear). Stan is a platform for statistical modeling and high-performance Bayesian inference using Hamiltonian Monte Carlo (HMC)¹¹ and its variants. Brms translates these models into Stan code and performs Markov Chain Monte Carlo (MCMC) sampling to estimate posterior distributions of all model parameters.

The model was fitted as the following form:

$$y_{ij} = \mu + u_j + \epsilon_{ij}$$

where μ is the overall mean, y_{ij} is the feature value (normalized clone size or TE/MP enrichment ratio) of the i clone in the j clonotype, u_j is the random effect of the j clonotype modeling inter-clonotypic (TCR-intrinsic) variance, and ϵ_{ij} captures intra-clonotypic (TCR-extrinsic) residual variance.

Posterior distributions of both variance components were directly sampled by Stan and used to derive 95% confidence intervals which are accounted for variation in clonotype shardeness (i.e., number of clones belonging to the same clonotype but found in distinct hosts). Clonotypes with lower shardeness are regularized via posterior shrinkage, whereby their estimates are partially pooled toward the overall mean, reducing overfitting.

Each model was run with four MCMC chains, each comprising 2000 iterations (including warm-up), and the adapt_delta parameter was set to 0.95 to promote stable convergence. Convergence diagnostics (trace plots and R-hat statistics) were inspected as part of standard model validation procedures.

To assess the statistical significance of the variance components inferred from the Bayesian model, we computed posterior probabilities for pairwise comparisons between TCR-extrinsic, TCR-intrinsic, and total variances. We evaluated the proportion of posterior samples in which one variance component exceeded another. The probability that TCR-extrinsic variance was greater than TCR-intrinsic variance was calculated as $P(\sigma^2 > u^2)$, where σ^2 is the estimated residual variance and u^2 is the estimated variance of the clonotype-level random intercepts. These comparisons provide an interpretable measure of relative magnitude with full uncertainty propagation. Probabilities close to 1 indicate strong Bayesian evidence for a consistent ordering across posterior samples.

Posterior probabilities for pairwise comparisons of variance components based on Bayesian modeling of clone size and TE/MP enrichment ratio are shown in Table 1. The comparisons reflect the likelihood that one variance component exceeds another, based on 4000 posterior samples from the hierarchical model.

Table 1: Posterior probabilities for pairwise comparisons of variance components

Feature	Comparison	Probability
Clone Size	TCR extrinsic > TCR intrinsic	1
Clone Size	Total > TCR extrinsic	1
Clone Size	Total > TCR intrinsic	1
TE/MP score	TCR extrinsic > TCR intrinsic	0.9965
TE/MP score	Total > TCR extrinsic	1
TE/MP score	Total > TCR intrinsic	1

Supplementary References

1. Zehn, D., Lee, S.Y., and Bevan, M.J. (2009). Complete but curtailed T-cell response to very low-affinity antigen. *Nature* 458, 211-214. 10.1038/nature07657.
2. Busch, D.H., Pilip, I.M., Vijn, S., and Pamer, E.G. (1998). Coordinate regulation of complex T cell populations responding to bacterial infection. *Immunity* 8, 353-362. 10.1016/s1074-7613(00)80540-3.

3. Heumos, L., Schaar, A.C., Lance, C., Litinetskaya, A., Drost, F., Zappia, L., Lucken, M.D., Strobl, D.C., Henao, J., Curion, F., et al. (2023). Best practices for single-cell analysis across modalities. *Nat Rev Genet* 24, 550-572. 10.1038/s41576-023-00586-w.
4. Wolf, F.A., Angerer, P., and Theis, F.J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol* 19, 15. 10.1186/s13059-017-1382-0.
5. Sturm, G., Szabo, T., Fotakis, G., Haider, M., Rieder, D., Trajanoski, Z., and Finotello, F. (2020). Scirpy: a Scanpy extension for analyzing single-cell T-cell receptor-sequencing data. *Bioinformatics* 36, 4817-4818. 10.1093/bioinformatics/btaa611.
6. Bernstein, N.J., Fong, N.L., Lam, I., Roy, M.A., Hendrickson, D.G., and Kelley, D.R. (2020). Solo: Doublet Identification in Single-Cell RNA-Seq via Semi-Supervised Deep Learning. *Cell Syst* 11, 95-101 e105. 10.1016/j.cels.2020.05.010.
7. Young, M.D., and Behjati, S. (2020). SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *Gigascience* 9. 10.1093/gigascience/giaa151.
8. Gayoso, A., Lopez, R., Xing, G., Boyeau, P., Valiollah Pour Amiri, V., Hong, J., Wu, K., Jayasuriya, M., Mehlman, E., Langevin, M., et al. (2022). A Python library for probabilistic analysis of single-cell omics data. *Nat Biotechnol* 40, 163-166. 10.1038/s41587-021-01206-w.
9. Plambeck, M., Kazeroonian, A., Loeffler, D., Kretschmer, L., Salinno, C., Schroeder, T., Busch, D.H., Flossdorf, M., and Buchholz, V.R. (2022). Heritable changes in division speed accompany the diversification of single T cell fate. *Proc Natl Acad Sci U S A* 119. 10.1073/pnas.2116260119.
10. Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software* 80, 1–28. 10.18637/jss.v080.i01.
11. Betancourt, M. (2017). A Conceptual Introduction to Hamiltonian Monte Carlo. arXiv. <https://doi.org/10.48550/arXiv.1701.02434>.